

codata

Table of Contents

codata	1
codata_cli	41
codata_version	43
codata_constant_type	44

NAME

codata – library for fundamental physical constants

LIBRARY

Codata library (*libcodata*, *-lcodata*)

SYNOPSIS

Fortran **use codata**

C **#include "codata.h"**

Python **import pycodata**

DESCRIPTION

codata is a Fortran library providing the fundamental physical constants published by the CODATA. It includes the 2022, 2018, 2014 and 2010 recommended values, allowing applications to reproduce calculations based on specific CODATA releases.

The constants are implemented as a derived type that carries: *the name, the value, the uncertainty, and the unit*. All the constants are provided as **double precision reals**. Long constant names can be aliased with shorter names.

It provides:

- CODATA 2022 constants
- CODATA 2018 constants
- CODATA 2014 constants
- CODATA 2010 constants
- Native Fortran API
- C interoperability API
- Python bindings

Fortran API

exposes the constants as derived type **codata_constant_type** which carries *the name, the value, the uncertainty, and the unit*.

C API

exposes a structure **codata_constant_type** that defines the same members as in Fortran i.e. *the name, the value, the uncertainty, and the unit*.

Python

exposes a dictionary with the keys being *the name, the value, the uncertainty, and the unit*.

NOTES

The latest values (2022) do not have the year as a suffix in their name whereas older values (2010, 2014, 2018) feature the year as a suffix. All constants are declared with the same variable name in Fortran, C, and Python.

- [1] P. J. Mohr, B. N. Taylor, and D. B. Newell,
CODATA Recommended Values of the Fundamental Physical Constants: 2010,
Review of Modern Physics, vol. 84, 2012.
- [2] P. J. Mohr, B. N. Taylor, and D. B. Newell,
CODATA Recommended Values of the Fundamental Physical Constants: 2014,
Journal of Physical and Chemical Reference Data, vol. 45, 2016.
- [3] P. J. Mohr, B. N. Taylor, and D. B. Newell,
CODATA Recommended Values of the Fundamental Physical Constants: 2018,
Review of Modern Physics, vol. 93, 2021.
- [4] P. Mohr, D. Newell, B. Taylor, and E. Tiesinga,
CODATA Recommended Values of the Fundamental Physical Constants: 2022,
Accessed: May 05, 2025. [Online]. Available:
<https://arxiv.org/abs/2409.03787>
- [5] P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga,
CODATA Recommended Values of the Fundamental Physical Constants: 2022,
Reviews of Modern Physics, vol. 97, no. 2, p. 25002, Apr. 2025,
doi: 10.1103/RevModPhys.97.025002.

EXAMPLE

Example in Fortran

```

program example_in_f
use codata
implicit none
print '(A)', '##### EXAMPLE IN FORTRAN #####'
print '(A)', '# VERSION'
print *, "version = ", version()
print '(A)', '# CONSTANTS'
print *, "c = ", SPEED_OF_LIGHT_IN_VACUUM%value
print '(A)', '# UNCERTAINTY'
print *, "u(c) = ", SPEED_OF_LIGHT_IN_VACUUM%uncertainty
print '(A)', '# OLDER VALUES'
print '(A, F23.16)', "Mu_2022(latest) = ", MOLAR_MASS_CONSTANT%value
print '(A, F23.16)', "Mu_2018 = ", MOLAR_MASS_CONSTANT_2018%value
print '(A, F23.16)', "Mu_2014 = ", MOLAR_MASS_CONSTANT_2014%value
print '(A, F23.16)', "Mu_2010 = ", MOLAR_MASS_CONSTANT_2010%value
end program

```

Example in C

```

#include <stdio.h>
#include "codata.h"

int main(void){
printf("%f", c);
printf("##### EXAMPLE IN C #####\n");
printf("%s\n", "# VERSION");
printf("version = %s\n", codata_version());
printf("%s\n", "# CONSTANTS");
printf("c = %f\n", SPEED_OF_LIGHT_IN_VACUUM.value);
printf("%s\n", "# UNCERTAINTY");
printf("u(c) = %f\n", SPEED_OF_LIGHT_IN_VACUUM.uncertainty);
printf("%s\n", "# OLDER VALUES");
printf("Mu_2022(latest) = %23.16f\n", MOLAR_MASS_CONSTANT.value);
printf("Mu_2018 = %23.16f\n", MOLAR_MASS_CONSTANT_2018.value);
}

```

```
printf("Mu_2014 = %23.16f\n", MOLAR_MASS_CONSTANT_2014.value);
printf("Mu_2010 = %23.16f\n", MOLAR_MASS_CONSTANT_2010.value);
return 0;
}
```

Example in Python

```
import sys
sys.path.insert(0, "../py/src/")
import pycodata
print("##### EXAMPLE IN PYTHON #####")
print("# VERSION")
print(f"version = {pycodata.__version__}")
print("# Constants")
print("c =", pycodata.SPEED_OF_LIGHT_IN_VACUUM["value"])
print("# UNCERTAINTY")
print("u(c) =", pycodata.SPEED_OF_LIGHT_IN_VACUUM["uncertainty"])
print("# OLDER VALUES")
print("Mu_2022 =", pycodata.MOLAR_MASS_CONSTANT["value"])
print("Mu_2018 =", pycodata.MOLAR_MASS_CONSTANT_2018["value"])
print("Mu_2014 =", pycodata.MOLAR_MASS_CONSTANT_2014["value"])
print("Mu_2010 =", pycodata.MOLAR_MASS_CONSTANT_2010["value"])
```

SEE ALSO

codata(3), codata_cli(1), codata_version(3), codata_constant_type(3)

CODATA 2022

List of available constants:

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO
- ALPHA_PARTICLE_MASS
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV
- ALPHA_PARTICLE_MASS_IN_U
- ALPHA_PARTICLE_MOLAR_MASS
- ALPHA_PARTICLE_PROTON_MASS_RATIO
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS
- ALPHA_PARTICLE_RMS_CHARGE_RADIUS
- ANGSTROM_STAR
- ATOMIC_MASS_CONSTANT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP

- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_ACTION
- ATOMIC_UNIT_OF_CHARGE
- ATOMIC_UNIT_OF_CHARGE_DENSITY
- ATOMIC_UNIT_OF_CURRENT
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM
- ATOMIC_UNIT_OF_ELECTRIC_FIELD
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM
- ATOMIC_UNIT_OF_ENERGY
- ATOMIC_UNIT_OF_FORCE
- ATOMIC_UNIT_OF_LENGTH
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY
- ATOMIC_UNIT_OF_MAGNETIZABILITY
- ATOMIC_UNIT_OF_MASS
- ATOMIC_UNIT_OF_MOMENTUM
- ATOMIC_UNIT_OF_PERMITTIVITY
- ATOMIC_UNIT_OF_TIME
- ATOMIC_UNIT_OF_VELOCITY
- AVOGADRO_CONSTANT
- BOHR_MAGNETON
- BOHR_MAGNETON_IN_EV_T
- BOHR_MAGNETON_IN_HZ_T
- BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- BOHR_MAGNETON_IN_K_T
- BOHR_RADIUS
- BOLTZMANN_CONSTANT
- BOLTZMANN_CONSTANT_IN_EV_K
- BOLTZMANN_CONSTANT_IN_HZ_K
- BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN

- CHARACTERISTIC_IMPEDANCE_OF_VACUUM
- CLASSICAL_ELECTRON_RADIUS
- COMPTON_WAVELENGTH
- CONDUCTANCE_QUANTUM
- CONVENTIONAL_VALUE_OF_AMPERE_90
- CONVENTIONAL_VALUE_OF_COULOMB_90
- CONVENTIONAL_VALUE_OF_FARAD_90
- CONVENTIONAL_VALUE_OF_HENRY_90
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT
- CONVENTIONAL_VALUE_OF_OHM_90
- CONVENTIONAL_VALUE_OF_VOLT_90
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT
- CONVENTIONAL_VALUE_OF_WATT_90
- COPPER_X_UNIT
- DEUTERON_ELECTRON_MAG_MOM_RATIO
- DEUTERON_ELECTRON_MASS_RATIO
- DEUTERON_G_FACTOR
- DEUTERON_MAG_MOM
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- DEUTERON_MASS
- DEUTERON_MASS_ENERGY_EQUIVALENT
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV
- DEUTERON_MASS_IN_U
- DEUTERON_MOLAR_MASS
- DEUTERON_NEUTRON_MAG_MOM_RATIO
- DEUTERON_PROTON_MAG_MOM_RATIO
- DEUTERON_PROTON_MASS_RATIO
- DEUTERON_RELATIVE_ATOMIC_MASS
- DEUTERON_RMS_CHARGE_RADIUS
- ELECTRON_CHARGE_TO_MASS_QUOTIENT
- ELECTRON_DEUTERON_MAG_MOM_RATIO
- ELECTRON_DEUTERON_MASS_RATIO
- ELECTRON_G_FACTOR
- ELECTRON_GYROMAG_RATIO
- ELECTRON_GYROMAG_RATIO_IN_MHZ_T
- ELECTRON_HELION_MASS_RATIO
- ELECTRON_MAG_MOM

- ELECTRON_MAG_MOM_ANOMALY
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- ELECTRON_MASS
- ELECTRON_MASS_ENERGY_EQUIVALENT
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- ELECTRON_MASS_IN_U
- ELECTRON_MOLAR_MASS
- ELECTRON_MUON_MAG_MOM_RATIO
- ELECTRON_MUON_MASS_RATIO
- ELECTRON_NEUTRON_MAG_MOM_RATIO
- ELECTRON_NEUTRON_MASS_RATIO
- ELECTRON_PROTON_MAG_MOM_RATIO
- ELECTRON_PROTON_MASS_RATIO
- ELECTRON_RELATIVE_ATOMIC_MASS
- ELECTRON_TAU_MASS_RATIO
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- ELECTRON_TRITON_MASS_RATIO
- ELECTRON_VOLT
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP
- ELECTRON_VOLT_HARTREE_RELATIONSHIP
- ELECTRON_VOLT_HERTZ_RELATIONSHIP
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP
- ELECTRON_VOLT_JOULE_RELATIONSHIP
- ELECTRON_VOLT_KELVIN_RELATIONSHIP
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP
- ELEMENTARY_CHARGE
- ELEMENTARY_CHARGE_OVER_H_BAR
- FARADAY_CONSTANT
- FERMI_COUPLING_CONSTANT
- FINE_STRUCTURE_CONSTANT
- FIRST_RADIATION_CONSTANT
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP
- HARTREE_ELECTRON_VOLT_RELATIONSHIP
- HARTREE_ENERGY

- HARTREE_ENERGY_IN_EV
- HARTREE_HERTZ_RELATIONSHIP
- HARTREE_INVERSE_METER_RELATIONSHIP
- HARTREE_JOULE_RELATIONSHIP
- HARTREE_KELVIN_RELATIONSHIP
- HARTREE_KILOGRAM_RELATIONSHIP
- HELION_ELECTRON_MASS_RATIO
- HELION_G_FACTOR
- HELION_MAG_MOM
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- HELION_MASS
- HELION_MASS_ENERGY_EQUIVALENT
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV
- HELION_MASS_IN_U
- HELION_MOLAR_MASS
- HELION_PROTON_MASS_RATIO
- HELION_RELATIVE_ATOMIC_MASS
- HELION_SHIELDING_SHIFT
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP
- HERTZ_ELECTRON_VOLT_RELATIONSHIP
- HERTZ_HARTREE_RELATIONSHIP
- HERTZ_INVERSE_METER_RELATIONSHIP
- HERTZ_JOULE_RELATIONSHIP
- HERTZ_KELVIN_RELATIONSHIP
- HERTZ_KILOGRAM_RELATIONSHIP
- HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133
- INVERSE_FINE_STRUCTURE_CONSTANT
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP
- INVERSE_METER_HARTREE_RELATIONSHIP
- INVERSE_METER_HERTZ_RELATIONSHIP
- INVERSE_METER_JOULE_RELATIONSHIP
- INVERSE_METER_KELVIN_RELATIONSHIP
- INVERSE_METER_KILOGRAM_RELATIONSHIP
- INVERSE_OF_CONDUCTANCE_QUANTUM
- JOSEPHSON_CONSTANT
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP

- JOULE_ELECTRON_VOLT_RELATIONSHIP
- JOULE_HARTREE_RELATIONSHIP
- JOULE_HERTZ_RELATIONSHIP
- JOULE_INVERSE_METER_RELATIONSHIP
- JOULE_KELVIN_RELATIONSHIP
- JOULE_KILOGRAM_RELATIONSHIP
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP
- KELVIN_ELECTRON_VOLT_RELATIONSHIP
- KELVIN_HARTREE_RELATIONSHIP
- KELVIN_HERTZ_RELATIONSHIP
- KELVIN_INVERSE_METER_RELATIONSHIP
- KELVIN_JOULE_RELATIONSHIP
- KELVIN_KILOGRAM_RELATIONSHIP
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP
- KILOGRAM_HARTREE_RELATIONSHIP
- KILOGRAM_HERTZ_RELATIONSHIP
- KILOGRAM_INVERSE_METER_RELATIONSHIP
- KILOGRAM_JOULE_RELATIONSHIP
- KILOGRAM_KELVIN_RELATIONSHIP
- LATTICE_PARAMETER_OF_SILICON
- LATTICE_SPACING_OF_IDEAL_SI_220
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA
- LUMINOUS EFFICACY
- MAG_FLUX_QUANTUM
- MOLAR_GAS_CONSTANT
- MOLAR_MASS_CONSTANT
- MOLAR_MASS_OF_CARBON_12
- MOLAR_PLANCK_CONSTANT
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA
- MOLAR_VOLUME_OF_SILICON
- MOLYBDENUM_X_UNIT
- MUON_COMPTON_WAVELENGTH
- MUON_ELECTRON_MASS_RATIO
- MUON_G_FACTOR
- MUON_MAG_MOM

- MUON_MAG_MOM_ANOMALY
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- MUON_MASS
- MUON_MASS_ENERGY_EQUIVALENT
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV
- MUON_MASS_IN_U
- MUON_MOLAR_MASS
- MUON_NEUTRON_MASS_RATIO
- MUON_PROTON_MAG_MOM_RATIO
- MUON_PROTON_MASS_RATIO
- MUON_TAU_MASS_RATIO
- NATURAL_UNIT_OF_ACTION
- NATURAL_UNIT_OF_ACTION_IN_EV_S
- NATURAL_UNIT_OF_ENERGY
- NATURAL_UNIT_OF_ENERGY_IN_MEV
- NATURAL_UNIT_OF_LENGTH
- NATURAL_UNIT_OF_MASS
- NATURAL_UNIT_OF_MOMENTUM
- NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C
- NATURAL_UNIT_OF_TIME
- NATURAL_UNIT_OF_VELOCITY
- NEUTRON_COMPTON_WAVELENGTH
- NEUTRON_ELECTRON_MAG_MOM_RATIO
- NEUTRON_ELECTRON_MASS_RATIO
- NEUTRON_G_FACTOR
- NEUTRON_GYROMAG_RATIO
- NEUTRON_GYROMAG_RATIO_IN_MHZ_T
- NEUTRON_MAG_MOM
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- NEUTRON_MASS
- NEUTRON_MASS_ENERGY_EQUIVALENT
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_MASS_IN_U
- NEUTRON_MOLAR_MASS
- NEUTRON_MUON_MASS_RATIO
- NEUTRON_PROTON_MAG_MOM_RATIO

- NEUTRON_PROTON_MASS_DIFFERENCE
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U
- NEUTRON_PROTON_MASS_RATIO
- NEUTRON_RELATIVE_ATOMIC_MASS
- NEUTRON_TAU_MASS_RATIO
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- NEWTONIAN_CONSTANT_OF_GRAVITATION
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C
- NUCLEAR_MAGNETON
- NUCLEAR_MAGNETON_IN_EV_T
- NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- NUCLEAR_MAGNETON_IN_K_T
- NUCLEAR_MAGNETON_IN_MHZ_T
- PLANCK_CONSTANT
- PLANCK_CONSTANT_IN_EV_HZ
- PLANCK_LENGTH
- PLANCK_MASS
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV
- PLANCK_TEMPERATURE
- PLANCK_TIME
- PROTON_CHARGE_TO_MASS_QUOTIENT
- PROTON_COMPTON_WAVELENGTH
- PROTON_ELECTRON_MASS_RATIO
- PROTON_G_FACTOR
- PROTON_GYROMAG_RATIO
- PROTON_GYROMAG_RATIO_IN_MHZ_T
- PROTON_MAG_MOM
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- PROTON_MAG_SHIELDING_CORRECTION
- PROTON_MASS
- PROTON_MASS_ENERGY_EQUIVALENT
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV
- PROTON_MASS_IN_U
- PROTON_MOLAR_MASS
- PROTON_MUON_MASS_RATIO

- PROTON_NEUTRON_MAG_MOM_RATIO
- PROTON_NEUTRON_MASS_RATIO
- PROTON_RELATIVE_ATOMIC_MASS
- PROTON_RMS_CHARGE_RADIUS
- PROTON_TAU_MASS_RATIO
- QUANTUM_OF_CIRCULATION
- QUANTUM_OF_CIRCULATION_TIMES_2
- REDUCED_COMPTON_WAVELENGTH
- REDUCED_MUON_COMPTON_WAVELENGTH
- REDUCED_NEUTRON_COMPTON_WAVELENGTH
- REDUCED_PLANCK_CONSTANT
- REDUCED_PLANCK_CONSTANT_IN_EV_S
- REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM
- REDUCED_PROTON_COMPTON_WAVELENGTH
- REDUCED_TAU_COMPTON_WAVELENGTH
- RYDBERG_CONSTANT
- RYDBERG_CONSTANT_TIMES_C_IN_HZ
- RYDBERG_CONSTANT_TIMES_HC_IN_EV
- RYDBERG_CONSTANT_TIMES_HC_IN_J
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA
- SECOND_RADIATION_CONSTANT
- SHIELDED_HELION_GYROMAG_RATIO
- SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_HELION_MAG_MOM
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- SHIELDED_PROTON_GYROMAG_RATIO
- SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_PROTON_MAG_MOM
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD
- SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT
- SPEED_OF_LIGHT_IN_VACUUM
- STANDARD_ACCELERATION_OF_GRAVITY

- STANDARD_ATMOSPHERE
- STANDARD_STATE_PRESSURE
- STEFAN_BOLTZMANN_CONSTANT
- TAU_COMPTON_WAVELENGTH
- TAU_ELECTRON_MASS_RATIO
- TAU_ENERGY_EQUIVALENT
- TAU_MASS
- TAU_MASS_ENERGY_EQUIVALENT
- TAU_MASS_IN_U
- TAU_MOLAR_MASS
- TAU_MUON_MASS_RATIO
- TAU_NEUTRON_MASS_RATIO
- TAU_PROTON_MASS_RATIO
- THOMSON_CROSS_SECTION
- TRITON_ELECTRON_MASS_RATIO
- TRITON_G_FACTOR
- TRITON_MAG_MOM
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- TRITON_MASS
- TRITON_MASS_ENERGY_EQUIVALENT
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV
- TRITON_MASS_IN_U
- TRITON_MOLAR_MASS
- TRITON_PROTON_MASS_RATIO
- TRITON_RELATIVE_ATOMIC_MASS
- TRITON_TO_PROTON_MAG_MOM_RATIO
- UNIFIED_ATOMIC_MASS_UNIT
- VACUUM_ELECTRIC_PERMITTIVITY
- VACUUM_MAG_PERMEABILITY
- VON_KLITZING_CONSTANT
- WEAK_MIXING_ANGLE
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT
- W_TO_Z_MASS_RATIO

CODATA 2018

List of available constants:

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO _2018

- ALPHA_PARTICLE_MASS_2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- ALPHA_PARTICLE_MASS_IN_U_2018
- ALPHA_PARTICLE_MOLAR_MASS_2018
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2018
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS_2018
- ANGSTROM_STAR_2018
- ATOMIC_MASS_CONSTANT_2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2018
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2018
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2018
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2018
- ATOMIC_UNIT_OF_ACTION_2018
- ATOMIC_UNIT_OF_CHARGE_2018
- ATOMIC_UNIT_OF_CHARGE_DENSITY_2018
- ATOMIC_UNIT_OF_CURRENT_2018
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2018
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2018
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2018
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2018
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2018
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2018
- ATOMIC_UNIT_OF_ENERGY_2018
- ATOMIC_UNIT_OF_FORCE_2018
- ATOMIC_UNIT_OF_LENGTH_2018
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2018
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2018
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2018
- ATOMIC_UNIT_OF_MASS_2018
- ATOMIC_UNIT_OF_MOMENTUM_2018

- ATOMIC_UNIT_OF_PERMITTIVITY _2018
- ATOMIC_UNIT_OF_TIME _2018
- ATOMIC_UNIT_OF_VELOCITY _2018
- AVOGADRO_CONSTANT _2018
- BOHR_MAGNETON _2018
- BOHR_MAGNETON_IN_EV_T _2018
- BOHR_MAGNETON_IN_HZ_T _2018
- BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA _2018
- BOHR_MAGNETON_IN_K_T _2018
- BOHR_RADIUS _2018
- BOLTZMANN_CONSTANT _2018
- BOLTZMANN_CONSTANT_IN_EV_K _2018
- BOLTZMANN_CONSTANT_IN_HZ_K _2018
- BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN _2018
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM _2018
- CLASSICAL_ELECTRON_RADIUS _2018
- COMPTON_WAVELENGTH _2018
- CONDUCTANCE_QUANTUM _2018
- CONVENTIONAL_VALUE_OF_AMPERE_90 _2018
- CONVENTIONAL_VALUE_OF_COULOMB_90 _2018
- CONVENTIONAL_VALUE_OF_FARAD_90 _2018
- CONVENTIONAL_VALUE_OF_HENRY_90 _2018
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT _2018
- CONVENTIONAL_VALUE_OF_OHM_90 _2018
- CONVENTIONAL_VALUE_OF_VOLT_90 _2018
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT _2018
- CONVENTIONAL_VALUE_OF_WATT_90 _2018
- COPPER_X_UNIT _2018
- DEUTERON_ELECTRON_MAG_MOM_RATIO _2018
- DEUTERON_ELECTRON_MASS_RATIO _2018
- DEUTERON_G_FACTOR _2018
- DEUTERON_MAG_MOM _2018
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2018
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2018
- DEUTERON_MASS _2018
- DEUTERON_MASS_ENERGY_EQUIVALENT _2018
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV _2018
- DEUTERON_MASS_IN_U _2018

- DEUTERON_MOLAR_MASS_2018
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2018
- DEUTERON_PROTON_MAG_MOM_RATIO_2018
- DEUTERON_PROTON_MASS_RATIO_2018
- DEUTERON_RELATIVE_ATOMIC_MASS_2018
- DEUTERON_RMS_CHARGE_RADIUS_2018
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2018
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2018
- ELECTRON_DEUTERON_MASS_RATIO_2018
- ELECTRON_G_FACTOR_2018
- ELECTRON_GYROMAG_RATIO_2018
- ELECTRON_GYROMAG_RATIO_IN_MHZ_T_2018
- ELECTRON_HELION_MASS_RATIO_2018
- ELECTRON_MAG_MOM_2018
- ELECTRON_MAG_MOM_ANOMALY_2018
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- ELECTRON_MASS_2018
- ELECTRON_MASS_ENERGY_EQUIVALENT_2018
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- ELECTRON_MASS_IN_U_2018
- ELECTRON_MOLAR_MASS_2018
- ELECTRON_MUON_MAG_MOM_RATIO_2018
- ELECTRON_MUON_MASS_RATIO_2018
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2018
- ELECTRON_NEUTRON_MASS_RATIO_2018
- ELECTRON_PROTON_MAG_MOM_RATIO_2018
- ELECTRON_PROTON_MASS_RATIO_2018
- ELECTRON_RELATIVE_ATOMIC_MASS_2018
- ELECTRON_TAU_MASS_RATIO_2018
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2018
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2018
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- ELECTRON_TRITON_MASS_RATIO_2018
- ELECTRON_VOLT_2018
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2018
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2018

- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP _2018
- ELECTRON_VOLT_JOULE_RELATIONSHIP _2018
- ELECTRON_VOLT_KELVIN_RELATIONSHIP _2018
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP _2018
- ELEMENTARY_CHARGE _2018
- ELEMENTARY_CHARGE_OVER_H_BAR _2018
- FARADAY_CONSTANT _2018
- FERMI_COUPLING_CONSTANT _2018
- FINE_STRUCTURE_CONSTANT _2018
- FIRST_RADIATION_CONSTANT _2018
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE _2018
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP _2018
- HARTREE_ELECTRON_VOLT_RELATIONSHIP _2018
- HARTREE_ENERGY _2018
- HARTREE_ENERGY_IN_EV _2018
- HARTREE_HERTZ_RELATIONSHIP _2018
- HARTREE_INVERSE_METER_RELATIONSHIP _2018
- HARTREE_JOULE_RELATIONSHIP _2018
- HARTREE_KELVIN_RELATIONSHIP _2018
- HARTREE_KILOGRAM_RELATIONSHIP _2018
- HELION_ELECTRON_MASS_RATIO _2018
- HELION_G_FACTOR _2018
- HELION_MAG_MOM _2018
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2018
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2018
- HELION_MASS _2018
- HELION_MASS_ENERGY_EQUIVALENT _2018
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV _2018
- HELION_MASS_IN_U _2018
- HELION_MOLAR_MASS _2018
- HELION_PROTON_MASS_RATIO _2018
- HELION_RELATIVE_ATOMIC_MASS _2018
- HELION_SHIELDING_SHIFT _2018
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP _2018
- HERTZ_ELECTRON_VOLT_RELATIONSHIP _2018
- HERTZ_HARTREE_RELATIONSHIP _2018
- HERTZ_INVERSE_METER_RELATIONSHIP _2018
- HERTZ_JOULE_RELATIONSHIP _2018

- HERTZ_KELVIN_RELATIONSHIP_2018
- HERTZ_KILOGRAM_RELATIONSHIP_2018
- HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133_2018
- INVERSE_FINE_STRUCTURE_CONSTANT_2018
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2018
- INVERSE_METER_HARTREE_RELATIONSHIP_2018
- INVERSE_METER_HERTZ_RELATIONSHIP_2018
- INVERSE_METER_JOULE_RELATIONSHIP_2018
- INVERSE_METER_KELVIN_RELATIONSHIP_2018
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2018
- INVERSE_OF_CONDUCTANCE_QUANTUM_2018
- JOSEPHSON_CONSTANT_2018
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- JOULE_ELECTRON_VOLT_RELATIONSHIP_2018
- JOULE_HARTREE_RELATIONSHIP_2018
- JOULE_HERTZ_RELATIONSHIP_2018
- JOULE_INVERSE_METER_RELATIONSHIP_2018
- JOULE_KELVIN_RELATIONSHIP_2018
- JOULE_KILOGRAM_RELATIONSHIP_2018
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2018
- KELVIN_HARTREE_RELATIONSHIP_2018
- KELVIN_HERTZ_RELATIONSHIP_2018
- KELVIN_INVERSE_METER_RELATIONSHIP_2018
- KELVIN_JOULE_RELATIONSHIP_2018
- KELVIN_KILOGRAM_RELATIONSHIP_2018
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2018
- KILOGRAM_HARTREE_RELATIONSHIP_2018
- KILOGRAM_HERTZ_RELATIONSHIP_2018
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2018
- KILOGRAM_JOULE_RELATIONSHIP_2018
- KILOGRAM_KELVIN_RELATIONSHIP_2018
- LATTICE_PARAMETER_OF_SILICON_2018
- LATTICE_SPACING_OF_IDEAL_SI_220_2018
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2018
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2018

- LUMINOUS_EFFICACY_2018
- MAG_FLUX_QUANTUM_2018
- MOLAR_GAS_CONSTANT_2018
- MOLAR_MASS_CONSTANT_2018
- MOLAR_MASS_OF_CARBON_12_2018
- MOLAR_PLANCK_CONSTANT_2018
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2018
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2018
- MOLAR_VOLUME_OF_SILICON_2018
- MOLYBDENUM_X_UNIT_2018
- MUON_COMPTON_WAVELENGTH_2018
- MUON_ELECTRON_MASS_RATIO_2018
- MUON_G_FACTOR_2018
- MUON_MAG_MOM_2018
- MUON_MAG_MOM_ANOMALY_2018
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- MUON_MASS_2018
- MUON_MASS_ENERGY_EQUIVALENT_2018
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- MUON_MASS_IN_U_2018
- MUON_MOLAR_MASS_2018
- MUON_NEUTRON_MASS_RATIO_2018
- MUON_PROTON_MAG_MOM_RATIO_2018
- MUON_PROTON_MASS_RATIO_2018
- MUON_TAU_MASS_RATIO_2018
- NATURAL_UNIT_OF_ACTION_2018
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2018
- NATURAL_UNIT_OF_ENERGY_2018
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2018
- NATURAL_UNIT_OF_LENGTH_2018
- NATURAL_UNIT_OF_MASS_2018
- NATURAL_UNIT_OF_MOMENTUM_2018
- NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C_2018
- NATURAL_UNIT_OF_TIME_2018
- NATURAL_UNIT_OF_VELOCITY_2018
- NEUTRON_COMPTON_WAVELENGTH_2018
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2018

- NEUTRON_ELECTRON_MASS_RATIO _2018
- NEUTRON_G_FACTOR _2018
- NEUTRON_GYROMAG_RATIO _2018
- NEUTRON_GYROMAG_RATIO_IN_MHZ_T _2018
- NEUTRON_MAG_MOM _2018
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2018
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2018
- NEUTRON_MASS _2018
- NEUTRON_MASS_ENERGY_EQUIVALENT _2018
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV _2018
- NEUTRON_MASS_IN_U _2018
- NEUTRON_MOLAR_MASS _2018
- NEUTRON_MUON_MASS_RATIO _2018
- NEUTRON_PROTON_MAG_MOM_RATIO _2018
- NEUTRON_PROTON_MASS_DIFFERENCE _2018
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT _2018
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV _2018
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U _2018
- NEUTRON_PROTON_MASS_RATIO _2018
- NEUTRON_RELATIVE_ATOMIC_MASS _2018
- NEUTRON_TAU_MASS_RATIO _2018
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO _2018
- NEWTONIAN_CONSTANT_OF_GRAVITATION _2018
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C _2018
- NUCLEAR_MAGNETON _2018
- NUCLEAR_MAGNETON_IN_EV_T _2018
- NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA _2018
- NUCLEAR_MAGNETON_IN_K_T _2018
- NUCLEAR_MAGNETON_IN_MHZ_T _2018
- PLANCK_CONSTANT _2018
- PLANCK_CONSTANT_IN_EV_HZ _2018
- PLANCK_LENGTH _2018
- PLANCK_MASS _2018
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV _2018
- PLANCK_TEMPERATURE _2018
- PLANCK_TIME _2018
- PROTON_CHARGE_TO_MASS_QUOTIENT _2018
- PROTON_COMPTON_WAVELENGTH _2018

- PROTON_ELECTRON_MASS_RATIO _2018
- PROTON_G_FACTOR _2018
- PROTON_GYROMAG_RATIO _2018
- PROTON_GYROMAG_RATIO_IN_MHZ_T _2018
- PROTON_MAG_MOM _2018
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2018
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2018
- PROTON_MAG_SHIELDING_CORRECTION _2018
- PROTON_MASS _2018
- PROTON_MASS_ENERGY_EQUIVALENT _2018
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV _2018
- PROTON_MASS_IN_U _2018
- PROTON_MOLAR_MASS _2018
- PROTON_MUON_MASS_RATIO _2018
- PROTON_NEUTRON_MAG_MOM_RATIO _2018
- PROTON_NEUTRON_MASS_RATIO _2018
- PROTON_RELATIVE_ATOMIC_MASS _2018
- PROTON_RMS_CHARGE_RADIUS _2018
- PROTON_TAU_MASS_RATIO _2018
- QUANTUM_OF_CIRCULATION _2018
- QUANTUM_OF_CIRCULATION_TIMES_2 _2018
- REDUCED_COMPTON_WAVELENGTH _2018
- REDUCED_MUON_COMPTON_WAVELENGTH _2018
- REDUCED_NEUTRON_COMPTON_WAVELENGTH _2018
- REDUCED_PLANCK_CONSTANT _2018
- REDUCED_PLANCK_CONSTANT_IN_EV_S _2018
- REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM _2018
- REDUCED_PROTON_COMPTON_WAVELENGTH _2018
- REDUCED_TAU_COMPTON_WAVELENGTH _2018
- RYDBERG_CONSTANT _2018
- RYDBERG_CONSTANT_TIMES_C_IN_HZ _2018
- RYDBERG_CONSTANT_TIMES_HC_IN_EV _2018
- RYDBERG_CONSTANT_TIMES_HC_IN_J _2018
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA _2018
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA _2018
- SECOND_RADIATION_CONSTANT _2018
- SHIELDED_HELION_GYROMAG_RATIO _2018
- SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T _2018

- SHIELDED_HELION_MAG_MOM_2018
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2018
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- SHIELDED_PROTON_GYROMAG_RATIO_2018
- SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T_2018
- SHIELDED_PROTON_MAG_MOM_2018
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD_2018
- SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT_2018
- SPEED_OF_LIGHT_IN_VACUUM_2018
- STANDARD_ACCELERATION_OF_GRAVITY_2018
- STANDARD_ATMOSPHERE_2018
- STANDARD_STATE_PRESSURE_2018
- STEFAN_BOLTZMANN_CONSTANT_2018
- TAU_COMPTON_WAVELENGTH_2018
- TAU_ELECTRON_MASS_RATIO_2018
- TAU_ENERGY_EQUIVALENT_2018
- TAU_MASS_2018
- TAU_MASS_ENERGY_EQUIVALENT_2018
- TAU_MASS_IN_U_2018
- TAU_MOLAR_MASS_2018
- TAU_MUON_MASS_RATIO_2018
- TAU_NEUTRON_MASS_RATIO_2018
- TAU_PROTON_MASS_RATIO_2018
- THOMSON_CROSS_SECTION_2018
- TRITON_ELECTRON_MASS_RATIO_2018
- TRITON_G_FACTOR_2018
- TRITON_MAG_MOM_2018
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- TRITON_MASS_2018
- TRITON_MASS_ENERGY_EQUIVALENT_2018
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- TRITON_MASS_IN_U_2018
- TRITON_MOLAR_MASS_2018

- TRITON_PROTON_MASS_RATIO _2018
- TRITON_RELATIVE_ATOMIC_MASS _2018
- TRITON_TO_PROTON_MAG_MOM_RATIO _2018
- UNIFIED_ATOMIC_MASS_UNIT _2018
- VACUUM_ELECTRIC_PERMITTIVITY _2018
- VACUUM_MAG_PERMEABILITY _2018
- VON_KLITZING_CONSTANT _2018
- WEAK_MIXING_ANGLE _2018
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT _2018
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT _2018
- W_TO_Z_MASS_RATIO _2018

CODATA 2014

List of available constants:

- LATTICE_SPACING_OF_SILICON _2014
- ALPHA_PARTICLE_ELECTRON_MASS_RATIO _2014
- ALPHA_PARTICLE_MASS _2014
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT _2014
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV _2014
- ALPHA_PARTICLE_MASS_IN_U _2014
- ALPHA_PARTICLE_MOLAR_MASS _2014
- ALPHA_PARTICLE_PROTON_MASS_RATIO _2014
- ANGSTROM_STAR _2014
- ATOMIC_MASS_CONSTANT _2014
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT _2014
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV _2014
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP _2014
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP _2014
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP _2014
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP _2014
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP _2014
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP _2014
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP _2014
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY _2014
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY _2014
- ATOMIC_UNIT_OF_ACTION _2014
- ATOMIC_UNIT_OF_CHARGE _2014
- ATOMIC_UNIT_OF_CHARGE_DENSITY _2014
- ATOMIC_UNIT_OF_CURRENT _2014

- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2014
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2014
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2014
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2014
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2014
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2014
- ATOMIC_UNIT_OF_ENERGY_2014
- ATOMIC_UNIT_OF_FORCE_2014
- ATOMIC_UNIT_OF_LENGTH_2014
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2014
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2014
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2014
- ATOMIC_UNIT_OF_MASS_2014
- ATOMIC_UNIT_OF_MOMUM_2014
- ATOMIC_UNIT_OF_PERMITTIVITY_2014
- ATOMIC_UNIT_OF_TIME_2014
- ATOMIC_UNIT_OF_VELOCITY_2014
- AVOGADRO_CONSTANT_2014
- BOHR_MAGNETON_2014
- BOHR_MAGNETON_IN_EV_T_2014
- BOHR_MAGNETON_IN_HZ_T_2014
- BOHR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2014
- BOHR_MAGNETON_IN_K_T_2014
- BOHR_RADIUS_2014
- BOLTZMANN_CONSTANT_2014
- BOLTZMANN_CONSTANT_IN_EV_K_2014
- BOLTZMANN_CONSTANT_IN_HZ_K_2014
- BOLTZMANN_CONSTANT_IN_INVERSE_METERS_PER_KELVIN_2014
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2014
- CLASSICAL_ELECTRON_RADIUS_2014
- COMPTON_WAVELENGTH_2014
- COMPTON_WAVELENGTH_OVER_2_PI_2014
- CONDUCTANCE_QUANTUM_2014
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2014
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2014
- CU_X_UNIT_2014
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2014
- DEUTERON_ELECTRON_MASS_RATIO_2014

- DEUTERON_G_FACTOR_2014
- DEUTERON_MAG_MOM_2014
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- DEUTERON_MASS_2014
- DEUTERON_MASS_ENERGY_EQUIVALENT_2014
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- DEUTERON_MASS_IN_U_2014
- DEUTERON_MOLAR_MASS_2014
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2014
- DEUTERON_PROTON_MAG_MOM_RATIO_2014
- DEUTERON_PROTON_MASS_RATIO_2014
- DEUTERON_RMS_CHARGE_RADIUS_2014
- ELECTRIC_CONSTANT_2014
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2014
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2014
- ELECTRON_DEUTERON_MASS_RATIO_2014
- ELECTRON_G_FACTOR_2014
- ELECTRON_GYROMAG_RATIO_2014
- ELECTRON_GYROMAG_RATIO_OVER_2_PI_2014
- ELECTRON_HELION_MASS_RATIO_2014
- ELECTRON_MAG_MOM_2014
- ELECTRON_MAG_MOM_ANOMALY_2014
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- ELECTRON_MASS_2014
- ELECTRON_MASS_ENERGY_EQUIVALENT_2014
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- ELECTRON_MASS_IN_U_2014
- ELECTRON_MOLAR_MASS_2014
- ELECTRON_MUON_MAG_MOM_RATIO_2014
- ELECTRON_MUON_MASS_RATIO_2014
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2014
- ELECTRON_NEUTRON_MASS_RATIO_2014
- ELECTRON_PROTON_MAG_MOM_RATIO_2014
- ELECTRON_PROTON_MASS_RATIO_2014
- ELECTRON_TAU_MASS_RATIO_2014
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2014

- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO _2014
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO _2014
- ELECTRON_TRITON_MASS_RATIO _2014
- ELECTRON_VOLT _2014
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP _2014
- ELECTRON_VOLT_HARTREE_RELATIONSHIP _2014
- ELECTRON_VOLT_HERTZ_RELATIONSHIP _2014
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP _2014
- ELECTRON_VOLT_JOULE_RELATIONSHIP _2014
- ELECTRON_VOLT_KELVIN_RELATIONSHIP _2014
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP _2014
- ELEMENTARY_CHARGE _2014
- ELEMENTARY_CHARGE_OVER_H _2014
- FARADAY_CONSTANT _2014
- FARADAY_CONSTANT_FOR_CONVENTIONAL_ELECTRIC_CURRENT _2014
- FERMI_COUPLING_CONSTANT _2014
- FINE_STRUCTURE_CONSTANT _2014
- FIRST_RADIATION_CONSTANT _2014
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE _2014
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP _2014
- HARTREE_ELECTRON_VOLT_RELATIONSHIP _2014
- HARTREE_ENERGY _2014
- HARTREE_ENERGY_IN_EV _2014
- HARTREE_HERTZ_RELATIONSHIP _2014
- HARTREE_INVERSE_METER_RELATIONSHIP _2014
- HARTREE_JOULE_RELATIONSHIP _2014
- HARTREE_KELVIN_RELATIONSHIP _2014
- HARTREE_KILOGRAM_RELATIONSHIP _2014
- HELION_ELECTRON_MASS_RATIO _2014
- HELION_G_FACTOR _2014
- HELION_MAG_MOM _2014
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2014
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2014
- HELION_MASS _2014
- HELION_MASS_ENERGY_EQUIVALENT _2014
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV _2014
- HELION_MASS_IN_U _2014
- HELION_MOLAR_MASS _2014

- HELION_PROTON_MASS_RATIO_2014
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2014
- HERTZ_HARTREE_RELATIONSHIP_2014
- HERTZ_INVERSE_METER_RELATIONSHIP_2014
- HERTZ_JOULE_RELATIONSHIP_2014
- HERTZ_KELVIN_RELATIONSHIP_2014
- HERTZ_KILOGRAM_RELATIONSHIP_2014
- INVERSE_FINE_STRUCTURE_CONSTANT_2014
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2014
- INVERSE_METER_HARTREE_RELATIONSHIP_2014
- INVERSE_METER_HERTZ_RELATIONSHIP_2014
- INVERSE_METER_JOULE_RELATIONSHIP_2014
- INVERSE_METER_KELVIN_RELATIONSHIP_2014
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2014
- INVERSE_OF_CONDUCTANCE_QUANTUM_2014
- JOSEPHSON_CONSTANT_2014
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- JOULE_ELECTRON_VOLT_RELATIONSHIP_2014
- JOULE_HARTREE_RELATIONSHIP_2014
- JOULE_HERTZ_RELATIONSHIP_2014
- JOULE_INVERSE_METER_RELATIONSHIP_2014
- JOULE_KELVIN_RELATIONSHIP_2014
- JOULE_KILOGRAM_RELATIONSHIP_2014
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2014
- KELVIN_HARTREE_RELATIONSHIP_2014
- KELVIN_HERTZ_RELATIONSHIP_2014
- KELVIN_INVERSE_METER_RELATIONSHIP_2014
- KELVIN_JOULE_RELATIONSHIP_2014
- KELVIN_KILOGRAM_RELATIONSHIP_2014
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2014
- KILOGRAM_HARTREE_RELATIONSHIP_2014
- KILOGRAM_HERTZ_RELATIONSHIP_2014
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2014
- KILOGRAM_JOULE_RELATIONSHIP_2014

- KILOGRAM_KELVIN_RELATIONSHIP_2014
- LATTICE_PARAMETER_OF_SILICON_2014
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2014
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2014
- MAG_CONSTANT_2014
- MAG_FLUX_QUANTUM_2014
- MOLAR_GAS_CONSTANT_2014
- MOLAR_MASS_CONSTANT_2014
- MOLAR_MASS_OF_CARBON_12_2014
- MOLAR_PLANCK_CONSTANT_2014
- MOLAR_PLANCK_CONSTANT_TIMES_C_2014
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2014
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2014
- MOLAR_VOLUME_OF_SILICON_2014
- MO_X_UNIT_2014
- MUON_COMPTON_WAVELENGTH_2014
- MUON_COMPTON_WAVELENGTH_OVER_2_PI_2014
- MUON_ELECTRON_MASS_RATIO_2014
- MUON_G_FACTOR_2014
- MUON_MAG_MOM_2014
- MUON_MAG_MOM_ANOMALY_2014
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- MUON_MASS_2014
- MUON_MASS_ENERGY_EQUIVALENT_2014
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- MUON_MASS_IN_U_2014
- MUON_MOLAR_MASS_2014
- MUON_NEUTRON_MASS_RATIO_2014
- MUON_PROTON_MAG_MOM_RATIO_2014
- MUON_PROTON_MASS_RATIO_2014
- MUON_TAU_MASS_RATIO_2014
- NATURAL_UNIT_OF_ACTION_2014
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2014
- NATURAL_UNIT_OF_ENERGY_2014
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2014
- NATURAL_UNIT_OF_LENGTH_2014
- NATURAL_UNIT_OF_MASS_2014

- NATURAL_UNIT_OF_MOMUM _2014
- NATURAL_UNIT_OF_MOMUM_IN_MEV_C _2014
- NATURAL_UNIT_OF_TIME _2014
- NATURAL_UNIT_OF_VELOCITY _2014
- NEUTRON_COMPTON_WAVELENGTH _2014
- NEUTRON_COMPTON_WAVELENGTH_OVER_2_PI _2014
- NEUTRON_ELECTRON_MAG_MOM_RATIO _2014
- NEUTRON_ELECTRON_MASS_RATIO _2014
- NEUTRON_G_FACTOR _2014
- NEUTRON_GYROMAG_RATIO _2014
- NEUTRON_GYROMAG_RATIO_OVER_2_PI _2014
- NEUTRON_MAG_MOM _2014
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2014
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2014
- NEUTRON_MASS _2014
- NEUTRON_MASS_ENERGY_EQUIVALENT _2014
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV _2014
- NEUTRON_MASS_IN_U _2014
- NEUTRON_MOLAR_MASS _2014
- NEUTRON_MUON_MASS_RATIO _2014
- NEUTRON_PROTON_MAG_MOM_RATIO _2014
- NEUTRON_PROTON_MASS_DIFFERENCE _2014
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT _2014
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV _2014
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U _2014
- NEUTRON_PROTON_MASS_RATIO _2014
- NEUTRON_TAU_MASS_RATIO _2014
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO _2014
- NEWTONIAN_CONSTANT_OF_GRAVITATION _2014
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C _2014
- NUCLEAR_MAGNETON _2014
- NUCLEAR_MAGNETON_IN_EV_T _2014
- NUCLEAR_MAGNETON_IN_INVERSE_METERS_PER_TESLA _2014
- NUCLEAR_MAGNETON_IN_K_T _2014
- NUCLEAR_MAGNETON_IN_MHZ_T _2014
- PLANCK_CONSTANT _2014
- PLANCK_CONSTANT_IN_EV_S _2014
- PLANCK_CONSTANT_OVER_2_PI _2014

- PLANCK_CONSTANT_OVER_2_PI_IN_EV_S_2014
- PLANCK_CONSTANT_OVER_2_PI_TIMES_C_IN_MEV_FM_2014
- PLANCK_LENGTH_2014
- PLANCK_MASS_2014
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2014
- PLANCK_TEMPERATURE_2014
- PLANCK_TIME_2014
- PROTON_CHARGE_TO_MASS_QUOTIENT_2014
- PROTON_COMPTON_WAVELENGTH_2014
- PROTON_COMPTON_WAVELENGTH_OVER_2_PI_2014
- PROTON_ELECTRON_MASS_RATIO_2014
- PROTON_G_FACTOR_2014
- PROTON_GYROMAG_RATIO_2014
- PROTON_GYROMAG_RATIO_OVER_2_PI_2014
- PROTON_MAG_MOM_2014
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- PROTON_MAG_SHIELDING_CORRECTION_2014
- PROTON_MASS_2014
- PROTON_MASS_ENERGY_EQUIVALENT_2014
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- PROTON_MASS_IN_U_2014
- PROTON_MOLAR_MASS_2014
- PROTON_MUON_MASS_RATIO_2014
- PROTON_NEUTRON_MAG_MOM_RATIO_2014
- PROTON_NEUTRON_MASS_RATIO_2014
- PROTON_RMS_CHARGE_RADIUS_2014
- PROTON_TAU_MASS_RATIO_2014
- QUANTUM_OF_CIRCULATION_2014
- QUANTUM_OF_CIRCULATION_TIMES_2_2014
- RYDBERG_CONSTANT_2014
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2014
- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2014
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2014
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2014
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2014
- SECOND_RADIATION_CONSTANT_2014
- SHIELDED_HELION_GYROMAG_RATIO_2014

- SHIELDED_HELION_GYROMAG_RATIO_OVER_2_PI_2014
- SHIELDED_HELION_MAG_MOM_2014
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2014
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014
- SHIELDED_PROTON_GYROMAG_RATIO_2014
- SHIELDED_PROTON_GYROMAG_RATIO_OVER_2_PI_2014
- SHIELDED_PROTON_MAG_MOM_2014
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- SPEED_OF_LIGHT_IN_VACUUM_2014
- STANDARD_ACCELERATION_OF_GRAVITY_2014
- STANDARD_ATMOSPHERE_2014
- STANDARD_STATE_PRESSURE_2014
- STEFAN_BOLTZMANN_CONSTANT_2014
- TAU_COMPTON_WAVELENGTH_2014
- TAU_COMPTON_WAVELENGTH_OVER_2_PI_2014
- TAU_ELECTRON_MASS_RATIO_2014
- TAU_MASS_2014
- TAU_MASS_ENERGY_EQUIVALENT_2014
- TAU_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- TAU_MASS_IN_U_2014
- TAU_MOLAR_MASS_2014
- TAU_MUON_MASS_RATIO_2014
- TAU_NEUTRON_MASS_RATIO_2014
- TAU_PROTON_MASS_RATIO_2014
- THOMSON_CROSS_SECTION_2014
- TRITON_ELECTRON_MASS_RATIO_2014
- TRITON_G_FACTOR_2014
- TRITON_MAG_MOM_2014
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- TRITON_MASS_2014
- TRITON_MASS_ENERGY_EQUIVALENT_2014
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- TRITON_MASS_IN_U_2014
- TRITON_MOLAR_MASS_2014

- TRITON_PROTON_MASS_RATIO _2014
- UNIFIED_ATOMIC_MASS_UNIT _2014
- VON_KLITZING_CONSTANT _2014
- WEAK_MIXING_ANGLE _2014
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT _2014
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT _2014

CODATA 2010

List of available constants:

- LATTICE_SPACING_OF_SILICON _2010
- ALPHA_PARTICLE_ELECTRON_MASS_RATIO _2010
- ALPHA_PARTICLE_MASS _2010
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT _2010
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV _2010
- ALPHA_PARTICLE_MASS_IN_U _2010
- ALPHA_PARTICLE_MOLAR_MASS _2010
- ALPHA_PARTICLE_PROTON_MASS_RATIO _2010
- ANGSTROM_STAR _2010
- ATOMIC_MASS_CONSTANT _2010
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT _2010
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV _2010
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP _2010
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP _2010
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP _2010
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP _2010
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP _2010
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP _2010
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP _2010
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY _2010
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY _2010
- ATOMIC_UNIT_OF_ACTION _2010
- ATOMIC_UNIT_OF_CHARGE _2010
- ATOMIC_UNIT_OF_CHARGE_DENSITY _2010
- ATOMIC_UNIT_OF_CURRENT _2010
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM _2010
- ATOMIC_UNIT_OF_ELECTRIC_FIELD _2010
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT _2010
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY _2010
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL _2010

- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM _2010
- ATOMIC_UNIT_OF_ENERGY _2010
- ATOMIC_UNIT_OF_FORCE _2010
- ATOMIC_UNIT_OF_LENGTH _2010
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM _2010
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY _2010
- ATOMIC_UNIT_OF_MAGNETIZABILITY _2010
- ATOMIC_UNIT_OF_MASS _2010
- ATOMIC_UNIT_OF_MOMUM _2010
- ATOMIC_UNIT_OF_PERMITTIVITY _2010
- ATOMIC_UNIT_OF_TIME _2010
- ATOMIC_UNIT_OF_VELOCITY _2010
- AVOGADRO_CONSTANT _2010
- BOHR_MAGNETON _2010
- BOHR_MAGNETON_IN_EV_T _2010
- BOHR_MAGNETON_IN_HZ_T _2010
- BOHR_MAGNETON_IN_INVERSE_METERS_PER_TESLA _2010
- BOHR_MAGNETON_IN_K_T _2010
- BOHR_RADIUS _2010
- BOLTZMANN_CONSTANT _2010
- BOLTZMANN_CONSTANT_IN_EV_K _2010
- BOLTZMANN_CONSTANT_IN_HZ_K _2010
- BOLTZMANN_CONSTANT_IN_INVERSE_METERS_PER_KELVIN _2010
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM _2010
- CLASSICAL_ELECTRON_RADIUS _2010
- COMPTON_WAVELENGTH _2010
- COMPTON_WAVELENGTH_OVER_2_PI _2010
- CONDUCTANCE_QUANTUM _2010
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT _2010
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT _2010
- CU_X_UNIT _2010
- DEUTERON_ELECTRON_MAG_MOM_RATIO _2010
- DEUTERON_ELECTRON_MASS_RATIO _2010
- DEUTERON_G_FACTOR _2010
- DEUTERON_MAG_MOM _2010
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2010
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2010
- DEUTERON_MASS _2010

- DEUTERON_MASS_ENERGY_EQUIVALENT _2010
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV _2010
- DEUTERON_MASS_IN_U _2010
- DEUTERON_MOLAR_MASS _2010
- DEUTERON_NEUTRON_MAG_MOM_RATIO _2010
- DEUTERON_PROTON_MAG_MOM_RATIO _2010
- DEUTERON_PROTON_MASS_RATIO _2010
- DEUTERON_RMS_CHARGE_RADIUS _2010
- ELECTRIC_CONSTANT _2010
- ELECTRON_CHARGE_TO_MASS_QUOTIENT _2010
- ELECTRON_DEUTERON_MAG_MOM_RATIO _2010
- ELECTRON_DEUTERON_MASS_RATIO _2010
- ELECTRON_G_FACTOR _2010
- ELECTRON_GYROMAG_RATIO _2010
- ELECTRON_GYROMAG_RATIO_OVER_2_PI _2010
- ELECTRON_HELION_MASS_RATIO _2010
- ELECTRON_MAG_MOM _2010
- ELECTRON_MAG_MOM_ANOMALY _2010
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2010
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2010
- ELECTRON_MASS _2010
- ELECTRON_MASS_ENERGY_EQUIVALENT _2010
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV _2010
- ELECTRON_MASS_IN_U _2010
- ELECTRON_MOLAR_MASS _2010
- ELECTRON_MUON_MAG_MOM_RATIO _2010
- ELECTRON_MUON_MASS_RATIO _2010
- ELECTRON_NEUTRON_MAG_MOM_RATIO _2010
- ELECTRON_NEUTRON_MASS_RATIO _2010
- ELECTRON_PROTON_MAG_MOM_RATIO _2010
- ELECTRON_PROTON_MASS_RATIO _2010
- ELECTRON_TAU_MASS_RATIO _2010
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO _2010
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO _2010
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO _2010
- ELECTRON_TRITON_MASS_RATIO _2010
- ELECTRON_VOLT _2010
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP _2010

- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2010
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2010
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2010
- ELECTRON_VOLT_JOULE_RELATIONSHIP_2010
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2010
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2010
- ELEMENTARY_CHARGE_2010
- ELEMENTARY_CHARGE_OVER_H_2010
- FARADAY_CONSTANT_2010
- FARADAY_CONSTANT_FOR_CONVENTIONAL_ELECTRIC_CURRENT_2010
- FERMI_COUPLING_CONSTANT_2010
- FINE_STRUCTURE_CONSTANT_2010
- FIRST_RADIATION_CONSTANT_2010
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2010
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- HARTREE_ELECTRON_VOLT_RELATIONSHIP_2010
- HARTREE_ENERGY_2010
- HARTREE_ENERGY_IN_EV_2010
- HARTREE_HERTZ_RELATIONSHIP_2010
- HARTREE_INVERSE_METER_RELATIONSHIP_2010
- HARTREE_JOULE_RELATIONSHIP_2010
- HARTREE_KELVIN_RELATIONSHIP_2010
- HARTREE_KILOGRAM_RELATIONSHIP_2010
- HELION_ELECTRON_MASS_RATIO_2010
- HELION_G_FACTOR_2010
- HELION_MAG_MOM_2010
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- HELION_MASS_2010
- HELION_MASS_ENERGY_EQUIVALENT_2010
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- HELION_MASS_IN_U_2010
- HELION_MOLAR_MASS_2010
- HELION_PROTON_MASS_RATIO_2010
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2010
- HERTZ_HARTREE_RELATIONSHIP_2010
- HERTZ_INVERSE_METER_RELATIONSHIP_2010

- HERTZ_JOULE_RELATIONSHIP_2010
- HERTZ_KELVIN_RELATIONSHIP_2010
- HERTZ_KILOGRAM_RELATIONSHIP_2010
- INVERSE_FINE_STRUCTURE_CONSTANT_2010
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2010
- INVERSE_METER_HARTREE_RELATIONSHIP_2010
- INVERSE_METER_HERTZ_RELATIONSHIP_2010
- INVERSE_METER_JOULE_RELATIONSHIP_2010
- INVERSE_METER_KELVIN_RELATIONSHIP_2010
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2010
- INVERSE_OF_CONDUCTANCE_QUANTUM_2010
- JOSEPHSON_CONSTANT_2010
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- JOULE_ELECTRON_VOLT_RELATIONSHIP_2010
- JOULE_HARTREE_RELATIONSHIP_2010
- JOULE_HERTZ_RELATIONSHIP_2010
- JOULE_INVERSE_METER_RELATIONSHIP_2010
- JOULE_KELVIN_RELATIONSHIP_2010
- JOULE_KILOGRAM_RELATIONSHIP_2010
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2010
- KELVIN_HARTREE_RELATIONSHIP_2010
- KELVIN_HERTZ_RELATIONSHIP_2010
- KELVIN_INVERSE_METER_RELATIONSHIP_2010
- KELVIN_JOULE_RELATIONSHIP_2010
- KELVIN_KILOGRAM_RELATIONSHIP_2010
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2010
- KILOGRAM_HARTREE_RELATIONSHIP_2010
- KILOGRAM_HERTZ_RELATIONSHIP_2010
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2010
- KILOGRAM_JOULE_RELATIONSHIP_2010
- KILOGRAM_KELVIN_RELATIONSHIP_2010
- LATTICE_PARAMETER_OF_SILICON_2010
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2010
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2010
- MAG_CONSTANT_2010

- MAG_FLUX_QUANTUM _2010
- MOLAR_GAS_CONSTANT _2010
- MOLAR_MASS_CONSTANT _2010
- MOLAR_MASS_OF_CARBON_12 _2010
- MOLAR_PLANCK_CONSTANT _2010
- MOLAR_PLANCK_CONSTANT_TIMES_C _2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA _2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA _2010
- MOLAR_VOLUME_OF_SILICON _2010
- MO_X_UNIT _2010
- MUON_COMPTON_WAVELENGTH _2010
- MUON_COMPTON_WAVELENGTH_OVER_2_PI _2010
- MUON_ELECTRON_MASS_RATIO _2010
- MUON_G_FACTOR _2010
- MUON_MAG_MOM _2010
- MUON_MAG_MOM_ANOMALY _2010
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2010
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2010
- MUON_MASS _2010
- MUON_MASS_ENERGY_EQUIVALENT _2010
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV _2010
- MUON_MASS_IN_U _2010
- MUON_MOLAR_MASS _2010
- MUON_NEUTRON_MASS_RATIO _2010
- MUON_PROTON_MAG_MOM_RATIO _2010
- MUON_PROTON_MASS_RATIO _2010
- MUON_TAU_MASS_RATIO _2010
- NATURAL_UNIT_OF_ACTION _2010
- NATURAL_UNIT_OF_ACTION_IN_EV_S _2010
- NATURAL_UNIT_OF_ENERGY _2010
- NATURAL_UNIT_OF_ENERGY_IN_MEV _2010
- NATURAL_UNIT_OF_LENGTH _2010
- NATURAL_UNIT_OF_MASS _2010
- NATURAL_UNIT_OF_MOMUM _2010
- NATURAL_UNIT_OF_MOMUM_IN_MEV_C _2010
- NATURAL_UNIT_OF_TIME _2010
- NATURAL_UNIT_OF_VELOCITY _2010
- NEUTRON_COMPTON_WAVELENGTH _2010

- NEUTRON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2010
- NEUTRON_ELECTRON_MASS_RATIO_2010
- NEUTRON_G_FACTOR_2010
- NEUTRON_GYROMAG_RATIO_2010
- NEUTRON_GYROMAG_RATIO_OVER_2_PI_2010
- NEUTRON_MAG_MOM_2010
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- NEUTRON_MASS_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- NEUTRON_MASS_IN_U_2010
- NEUTRON_MOLAR_MASS_2010
- NEUTRON_MUON_MASS_RATIO_2010
- NEUTRON_PROTON_MAG_MOM_RATIO_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2010
- NEUTRON_PROTON_MASS_RATIO_2010
- NEUTRON_TAU_MASS_RATIO_2010
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010
- NEWTONIAN_CONSTANT_OF_GRAVITATION_2010
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2010
- NUCLEAR_MAGNETON_2010
- NUCLEAR_MAGNETON_IN_EV_T_2010
- NUCLEAR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2010
- NUCLEAR_MAGNETON_IN_K_T_2010
- NUCLEAR_MAGNETON_IN_MHZ_T_2010
- PLANCK_CONSTANT_2010
- PLANCK_CONSTANT_IN_EV_S_2010
- PLANCK_CONSTANT_OVER_2_PI_2010
- PLANCK_CONSTANT_OVER_2_PI_IN_EV_S_2010
- PLANCK_CONSTANT_OVER_2_PI_TIMES_C_IN_MEV_FM_2010
- PLANCK_LENGTH_2010
- PLANCK_MASS_2010
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2010

- PLANCK_TEMPERATURE _2010
- PLANCK_TIME _2010
- PROTON_CHARGE_TO_MASS_QUOTIENT _2010
- PROTON_COMPTON_WAVELENGTH _2010
- PROTON_COMPTON_WAVELENGTH_OVER_2_PI _2010
- PROTON_ELECTRON_MASS_RATIO _2010
- PROTON_G_FACTOR _2010
- PROTON_GYROMAG_RATIO _2010
- PROTON_GYROMAG_RATIO_OVER_2_PI _2010
- PROTON_MAG_MOM _2010
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2010
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2010
- PROTON_MAG_SHIELDING_CORRECTION _2010
- PROTON_MASS _2010
- PROTON_MASS_ENERGY_EQUIVALENT _2010
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV _2010
- PROTON_MASS_IN_U _2010
- PROTON_MOLAR_MASS _2010
- PROTON_MUON_MASS_RATIO _2010
- PROTON_NEUTRON_MAG_MOM_RATIO _2010
- PROTON_NEUTRON_MASS_RATIO _2010
- PROTON_RMS_CHARGE_RADIUS _2010
- PROTON_TAU_MASS_RATIO _2010
- QUANTUM_OF_CIRCULATION _2010
- QUANTUM_OF_CIRCULATION_TIMES_2 _2010
- RYDBERG_CONSTANT _2010
- RYDBERG_CONSTANT_TIMES_C_IN_HZ _2010
- RYDBERG_CONSTANT_TIMES_HC_IN_EV _2010
- RYDBERG_CONSTANT_TIMES_HC_IN_J _2010
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA _2010
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA _2010
- SECOND_RADIATION_CONSTANT _2010
- SHIELDED_HELION_GYROMAG_RATIO _2010
- SHIELDED_HELION_GYROMAG_RATIO_OVER_2_PI _2010
- SHIELDED_HELION_MAG_MOM _2010
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2010
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2010
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO _2010

- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO _2010
- SHIELDED_PROTON_GYROMAG_RATIO _2010
- SHIELDED_PROTON_GYROMAG_RATIO_OVER_2_PI _2010
- SHIELDED_PROTON_MAG_MOM _2010
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2010
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2010
- SPEED_OF_LIGHT_IN_VACUUM _2010
- STANDARD_ACCELERATION_OF_GRAVITY _2010
- STANDARD_ATMOSPHERE _2010
- STANDARD_STATE_PRESSURE _2010
- STEFAN_BOLTZMANN_CONSTANT _2010
- TAU_COMPTON_WAVELENGTH _2010
- TAU_COMPTON_WAVELENGTH_OVER_2_PI _2010
- TAU_ELECTRON_MASS_RATIO _2010
- TAU_MASS _2010
- TAU_MASS_ENERGY_EQUIVALENT _2010
- TAU_MASS_ENERGY_EQUIVALENT_IN_MEV _2010
- TAU_MASS_IN_U _2010
- TAU_MOLAR_MASS _2010
- TAU_MUON_MASS_RATIO _2010
- TAU_NEUTRON_MASS_RATIO _2010
- TAU_PROTON_MASS_RATIO _2010
- THOMSON_CROSS_SECTION _2010
- TRITON_ELECTRON_MASS_RATIO _2010
- TRITON_G_FACTOR _2010
- TRITON_MAG_MOM _2010
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2010
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2010
- TRITON_MASS _2010
- TRITON_MASS_ENERGY_EQUIVALENT _2010
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV _2010
- TRITON_MASS_IN_U _2010
- TRITON_MOLAR_MASS _2010
- TRITON_PROTON_MASS_RATIO _2010
- UNIFIED_ATOMIC_MASS_UNIT _2010
- VON_KLITZING_CONSTANT _2010
- WEAK_MIXING_ANGLE _2010
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT _2010

- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2010

NAME

codata – library for fundamental physical constants

SYNOPSIS

codata [*OPTION*]...

DESCRIPTION

codata is a command line interface which prints all the CODATA constants.

The current values are from 2022. Older values can be retrieved if needed and the output can be filtered with *REGEX PATTERNS*.

OPTIONS

- y, --year *YEAR***
CODATA constants: 2022, 2018, 2014, 2010
- p, --pattern *PATTERN***
Regex pattern for filtering the constants.
- a, --value**
Show only the value.
- e, --error**
Show only the uncertainty.
- u, --usage**
Show usage text and exit.
- v, --version**
Show version information and exit.
- h, --help**
Show help text and exit.

NOTES

You may replace the default options from a file if your first options begin with @file. Initial options will then be read from the "response file" "file.rsp" in the current directory.

If "file" does not exist or cannot be read, then an error occurs and the program stops. Each line of the file is prefixed with "options" and interpreted as a separate argument. The file itself may not contain @file arguments. That is, it is not processed recursively.

For more information on response files see https://urbanjost.github.io/M_CLI2/set_args.3m_cli2.html

EXAMPLE

```
Minimal example
codata
codata -p Faraday
```

```
codata -y 2014 -p molar.*gas,electron.*eV
codata [B,b]oltzmann.*eV
```

SEE ALSO

codata(3), codata_cli(1), codata_version(3), codata_constant_type(3)

NAME

version – function

LIBRARY

Codata library (*libcodata*, *-lcodata*)

SYNOPSIS

Fortran **character(len=:), pointer :: version()**

C **char* codata_version()**

Python **pycodata.__version__**

DESCRIPTION

Get the library version.

EXAMPLE

Fortran:

```
use codata
print *, "version = ", version()
```

C:

```
include <stdio.h>
include "codata.h"
printf("version = %s0, codata_version());
```

Python:

```
import pycodata
print(f"version = {pycodata.__version__}")
```

SEE ALSO

codata(3), codata_cli(1), codata_version(3), codata_constant_type(3)

NAME

codata_constant_type – derived type

LIBRARY

Codata library (*libcodata*, *-lcodata*)

SYNOPSIS

Fortran **type(codata_constant_type) :: c**

C **struct codata_constant_type c**

DESCRIPTION

Defines a constant with *the name, the value, the uncertainty and the unit*.

```

type :: codata_constant_type
    character(len=64) :: name
    real(dp) :: value
    real(dp) :: uncertainty
    character(len=32) :: unit
contains
    procedure :: print
    procedure :: to_real_sp
    procedure :: to_real_dp
    generic :: to_real => to_real_sp, to_real_dp
end type codata_constant_type

struct codata_constant_type{
    char name[65];
    double value;
    double uncertainty;
    char unit[33];
};

```

EXAMPLE

Fortran:

```

use codata
type(codata_constant_type), pointer :: c
c = SPEED_OF_LIGHT_IN_VACUUM

```

C:

```

include "codata.h"
struct codata_constant_type c;
c = SPEED_OF_LIGHT_IN_VACUUM;

```

SEE ALSO**codata(3), codata_cli(1), codata_version(3), codata_constant_type(3)**